

**NIRMALA MEMORIAL FOUNDATION COLLEGE OF
COMMERCE AND SCIENCE
(AUTONOMOUS)**

Re-Accredited by NAAC with B++ Grade & ISO 9001:2015 Certified Recognized under
Section 2 (f) & 12 (B) of the UGC Act, 1956.
Kandivali (East), Mumbai-400101.



**IoT-Based Bus Stop System for Public Transport
Overcrowding Prediction Using Machine Learning**

A Project Report

Submitted in partial fulfillment of the requirements
For the degree of B.Sc. Information Technology aligned with
the NEP-2020 Community Engagement Project (CEP)

By

Mr. Jashith Agre and Mr. Om Mangesh Keluskar

Seat No: 24TIT011, 24TIT078

Semester: V

Under the esteemed guidance of

Mr. George Thekkevilayil

Assistant Professor

BACHELOR OF SCIENCE INFORMATION TECHNOLOGY

Academic Year: 2026-2027



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D S Road, Asha Nagar, Thakur Complex, Kandivali (East), Mumbai-400101 Tel:022-69436400

CERTIFICATE OF PROJECT COMPLETION

B.Sc. (Information Technology)

This is to certify that **Mr. Jashith Agre** and **Mr. Om Mangesh Keluskar**, students of T.Y.B.Sc. (Information Technology), Semester V, bearing Seat No. 24TIT011 and 24TIT078, has successfully completed the Project Work titled "**IoT-Based Bus Stop System for Public Transport Overcrowding Prediction Using Machine Learning.**" The project has been submitted in partial fulfillment of the requirements prescribed in the syllabus for the award of the Degree of **Bachelor of Science in Information Technology** for the academic year **2026–2027**. The practical and project work has been carried out successfully under the supervision of the undersigned.

Internal Examiner

Prof. George Thekkevilayil

External Examiner

Co-ordinator

Prof. Vaishali Mishra

College seal and Date



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PROJECT PROPOSAL APPROVAL SHEET

Department of Information Technology
Academic Year 2026-27

1. Name of the Students: Mr. Jashith Agre and Mr. Om Mangesh Keluskar

2. Seat Number: 24TIT011, 24TIT078

3. Is this your first submission? Yes ☐ No ☐

4. Name of the Guide: Prof. George Thekkevilayil

5. Teaching Experience of the Guide: 26 Years

The proposed project titled "**Iot-Based Bus Stop System for Public Transport Overcrowding Prediction Using Machine Learning.**" submitted by the above students, has been examined and found suitable for the partial fulfilment of the requirements for **Semester V** of the B.Sc IT Degree Programme. The project is hereby approved for implementation under the guidance of the undersigned.

Signature of Guide

Date:

Signature of Co-ordinator

Date:



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**Department of Information Technology
Community Engagement Project Questionnaire**

Name of Student(s): Mr.Jashith Agre, Mr.Om Mangesh Keluskar

Seat No: 24TIT011,24TIT078

Name of Participant:

Signature of Participant

Survey Questionnaire

1. How frequently do you experience overcrowding at Dindoshi Bus Depot or the BEST Route 326 bus stop? ☐ Every Day ☐ Several Times a Week ☐ Occasionally ☐ Rarely
2. Which mode of transport do you use most frequently to reach this stop? ☐ BEST bus (Route 326) ☐ Other BEST Bus ☐ Train then BEST Bus ☐ Walking
3. What is the biggest problem you face because of overcrowding at the bus stop? ☐ Missed bus / cannot board ☐ Pushing or injury risk ☐ Long uncomfortable wait ☐ No crowding warning
4. Have you ever been unable to board a bus because the stop or doorway was too crowded? ☐ Yes, Multiple Times ☐ Yes, Once ☐ No ☐ Not Applicable
5. How safe do you feel while waiting at an overcrowded bus stop? ☐ Very Safe ☐ Safe ☐ Unsafe ☐ Very Unsafe
6. Have you ever reported overcrowding to depot staff or BEST officials? ☐ Yes ☐ No ☐ Didn't Know How ☐ Never Felt It Would Help
7. If you did report overcrowding, how satisfied were you with the response? ☐ Very Satisfied ☐ Satisfied ☐ Dissatisfied ☐ Never Reported
8. How important is it to warn commuters as soon as overcrowding is detected? ☐ Extremely Important ☐ Important ☐ Slightly Important ☐ Not Important
9. Do you think rain, heat, or peak hours make overcrowding at the stop more dangerous? ☐ Strongly Agree ☐ Agree ☐ Disagree ☐ Strongly Disagree
10. Would you trust an automated IoT system that predicts overcrowding, without requiring a mobile app? ☐ Definitely Yes ☐ Probably Yes ☐ Probably No ☐ Definitely No
11. Which feature would be most useful in a smart bus-stop overcrowding system? ☐ On-site LED display ☐ Telegram staff alerts ☐ Live risk percentage ☐ Bus-left operator button
12. Do you believe an IoT-based pothole detection system can improve road safety and municipal maintenance? ☐ Strongly Agree ☐ Agree ☐ Disagree ☐ Strongly Disagree



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**Department of Information Technology
Community Engagement Project Questionnaire**

Name of Student(s): Mr.Jashith Agre, Mr.Om Mangesh Keluskar

Seat No: 24TIT011,24TIT078

Name of Participant:

Signature of Participant

Feedback Questionnaire

1. How easy was it to understand the purpose of the IoT bus-stop overcrowding system? ☐ VeryEasy ☐ Easy ☐ Neutral ☐ Difficult
2.How useful do you think automatic passenger counting using the IoT sensor would be for overcrowding awareness? ☐ VeryUseful ☐ Useful ☐ ModeratelyUseful ☐ Not Useful at All
3.How clear and understandable was the demonstrated overcrowding-prediction process? ☐ VeryClear ☐ Clear ☐ Neutral ☐ Unclear
4.How useful were the on-site red and green LED indicators for understanding crowding? ☐ Useful ☐ ModeratelyUseful ☐ SlightlyUseful ☐ Not Useful at All
5.How useful would real-time Telegram alerts be for depot staff when crowding becomes CRITICAL? ☐ VeryEasy ☐ Easy ☐ Neutral ☐ Difficult
6.How easy was it to understand the crowding information shown on the dashboard (count, weather, risk, status)? ☐ VeryEasy ☐ Easy ☐ Neutral ☐ VeryDifficult
7.How clearly did the hardware demonstration show how the sensor detects passenger crossings? ☐ VeryClear ☐ Clear ☐ Neutral ☐ Unclear
8.How confident are you that this system can warn of overcrowding faster than manual visual guessing? ☐ VeryConfident ☐ Confident ☐ ModeratelyConfident ☐ SlightlyConfident
9.Would you be willing to use this system if it remained available at Dindoshi Route 326? ☐ DefinitelyYes ☐ ProbablyYes ☐ Not Sure ☐ DefinitelyNo
10.How clear and easy to understand were the NORMAL, WARNING, and CRITICAL status lights? ☐ VeryClear ☐ Clear ☐ Neutral ☐ Unclear
11.How useful do you think this system would be for improving crowding awareness at your bus stop? ☐ VeryUseful ☐ Useful ☐ ModeratelyUseful ☐ Not Useful at All
12.Overall, how satisfied are you with the IoT bus-stop overcrowding system and its potential for commuter safety? ☐ VerySatisfied ☐ Satisfied ☐ Neutral ☐ Dissatisfied